



Hello everyone and wel...
TXT 18.87KB

explain everything properly

Machine Learning Foundations: Classification Learning Explained

This lecture covers **classification** in supervised learning, building on the regression problem from the previous lecture. Let me break down all the key concepts clearly.

1. What is Classification?

The Core Idea

In **regression**, you predict a real-valued number (e.g., house price = \$350,000).

In **classification**, you predict a **discrete label** — one of a finite set of categories.

The Running Example

Predict whether a house has **more than 3 rooms** or **≤ 3 rooms**, based on features like area and price.

- Output is binary: $+1$ (more than 3 rooms) or -1 (3 or fewer rooms)
- This is a **binary classification** problem

2. Training Data for Classification

Aspect	Regression	Classification
Inputs (x_i)	d -dimensional vectors ($x_i \in \mathbb{R}^d$)	Same — $x_i \in \mathbb{R}^d$
Labels (y_i)	Real numbers ($y_i \in \mathbb{R}$)	Discrete: $y_i \in \{+1, -1\}$
Model output	$f: \mathbb{R}^d \rightarrow \mathbb{R}$	$f: \mathbb{R}^d \rightarrow \{+1, -1\}$

Key difference: The labels are no longer real-valued — they're just $+1$ or -1 .

3. Evaluating a Classification Model

Loss Function

The **loss** of a model f is the **fraction of misclassified instances**:

$$\text{Loss}(f) = \frac{1}{n} \sum_{i=1}^n \mathbb{1}[f(x_i) \neq y_i]$$

Where the **indicator function** $\mathbb{1}[\text{condition}]$ equals:

- **1** if the condition is true (misclassified)
- **0** if the condition is false (correctly classified)

Ideal Case

Perfect model: $f(x_i) = y_i$ for **all** $i \rightarrow \text{Loss} = 0$

4. Linear Classifier (Linear Separator)

Parameterization

For regression, we used: $f(x) = w^T x + b$

For classification, we apply a **sign function** on top:

$$f(x) = \text{sign}(w^T x + b)$$

This is called a **linear separator** because it divides the input space into two regions with a straight line (or hyperplane).

5. Detailed 2D Example

The Data

Six 2D points:

Point	x_1	x_2	Label
1	0	0	+1
2	1	0	+1
3	0	1	+1

Point	x_1	x_2	Label
4	4	4	-1
5	3	4	-1
6	4	3	-1

Visualization:

- Positive points (red): clustered near origin
- Negative points (blue): clustered far away

Two Candidate Models

Model f: $f(x) = \text{sign}(2 - x_1)$

Model g: $g(x) = \text{sign}(x_1 - 2x_2)$

Evaluating Model f

Point	$2 - x_1$	sign	True y	Correct?
(0,0)	2	+1	+1	✓
(1,0)	1	+1	+1	✓
(0,1)	2	+1	+1	✓
(4,4)	-2	-1	-1	✓
(3,4)	-1	-1	-1	✓
(4,3)	-2	-1	-1	✓

Loss(f) = 0/6 = 0

Evaluating Model g

Point	$x_1 - 2x_2$	sign	True y	Correct?
(0,0)	0	+1*	+1	✓
(1,0)	1	+1	+1	✓
(0,1)	-2	-1	+1	X

Point	$x_1 - 2x_2$	sign	True y	Correct?
(4,4)	-4	-1	-1	✓
(3,4)	-5	-1	-1	✓
(4,3)	-2	-1	-1	✓

*Convention: $\text{sign}(0) = +1$

Loss(g) = 1/6

Conclusion

The learning algorithm prefers **f** over **g** because **f** has lower loss.

Visual Interpretation

- **f** draws a vertical line at $x_1 = 2$, perfectly separating positives from negatives
- **g** draws a line that misclassifies point (0,1)

6. Real-World Example: House Rooms Prediction

Setup

Predict if a house has > 3 rooms based on **area** and **price**.

Encoding:

- Rooms $\leq 3 \rightarrow$ label -1
- Rooms $> 3 \rightarrow$ label +1

Data (6 houses)

House	Area	Price	True Label
1	< 10	< 6	-1
2	< 10	< 6	-1
3	> 10	< 6	+1
4	> 10	> 9	+1
5	> 10	< 9	+1

House	Area	Price	True Label
6	> 10	< 9	+1

Three Candidate Models

$f(x) = \text{sign}(\text{area} - 10) \rightarrow$ Predicts +1 if area > 10

Predictions: [-1, -1, +1, +1, +1, +1] $\rightarrow$ **Loss = 0**

$g(x) = \text{sign}(\text{price} - 6) \rightarrow$ Predicts +1 if price > 6

Predictions: [-1, -1, +1, +1, +1, +1] $\rightarrow$ **Loss = 0**

$h(x) = \text{sign}(\text{price} - 9) \rightarrow$ Predicts +1 if price > 9

Predictions: [-1, -1, -1, +1, -1, -1] $\rightarrow$ **Loss = 3/6**

Conclusion

- **f and g** are both perfect on training data (loss = 0)
- The algorithm is **stumped** — it can pick either f or g
- **h** is rejected (loss = 3/6)

7. Why We Don't Evaluate on Training Data

The Overfitting Problem

Consider this "cheating" model:

$$f(x) = \begin{cases} +1 & \text{if } x = (0, 0), (1, 0), \text{ or } (0, 1) \\ -1 & \text{otherwise} \end{cases}$$

This model:

- Achieves **0 loss** on training data (memorizes the 3 positive points)
- Predicts -1 for **almost all** other inputs
- Will perform **terribly** on new data

The Real Goal

Machine learning is **not** about doing well on data you've already seen. It's about:

- Learning general principles

- Performing well on **unseen** future data

The Analogy

Using house prices from 1990–2020 to learn a model:

- Goal: Predict prices for houses in **2021**
- Predicting 1990 prices accurately is useless — you already know them

8. Train / Validation / Test Split

Dataset	Purpose	When Used
Training data	Find the best model within a chosen collection	During learning
Validation data	Choose the best collection/parameterization of models	During model selection
Test data	Evaluate the final learned model's performance	After learning is complete

Model Selection Explained

Who chooses the model class? Humans, based on intuition.

Example — two possible parameterizations:

1. $\text{price} = a \cdot \text{area} + b \cdot \text{rooms} + c \cdot \text{distance}$
2. $\text{price} = a \cdot (\text{area}/\text{rooms}) + b \cdot \text{distance}^2 + c \cdot (\text{distance}/\text{area})$

There's no mathematical reason to prefer one over the other — this is a **human design choice**.

Process:

1. Try both collections
2. Evaluate each on **validation data**
3. Pick the collection with better validation performance
4. Within that collection, find the best model using **training data**
5. Evaluate the final model on **test data**

9. Summary of Key Takeaways

1. **Classification** predicts discrete labels (+1/-1), unlike regression which predicts real numbers.
2. **Loss** = fraction of misclassified instances.
3. **Linear classifier**: $f(x) = \text{sign}(w^T x + b)$ — divides space with a hyperplane.
4. **Never evaluate on training data** — you'll get an overoptimistic estimate.
5. **Three data splits**:
 - Training → learn the model
 - Validation → choose the model class
 - Test → evaluate final performance
6. **Multiple models can have equal training loss** — the algorithm may be indifferent between them.
7. **The real goal** is generalization: performing well on unseen data, not memorizing training data.

10. What's Next

This concludes **supervised learning** (regression + classification). The next lecture will introduce **unsupervised learning**, where the data has no labels.